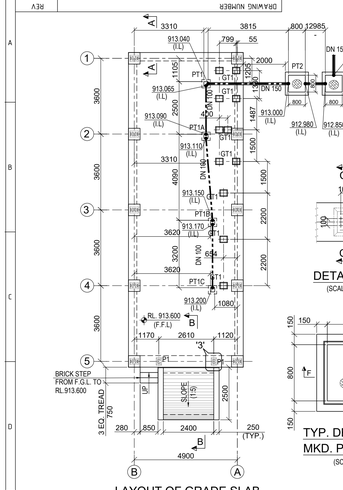
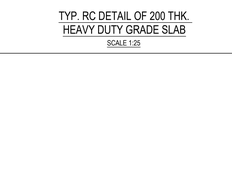
Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**GENERAL ARRANGEMENT POWER PLANT MISCELLANEOUS STRUCTURE G.A. AND DETAIL**
Document Number:	**MD-216-49PR-EG-CV-GAD-2024**
Revision:	**D01**
Revision Date:	**17.02.2023**
Status:	**AS BUILT**

Outotec Document Number:	900707	GAU01	WG01	MTL01	45092	OU100690015	7
L&T Document Number:	O19045	008	C	G	24298	006	7

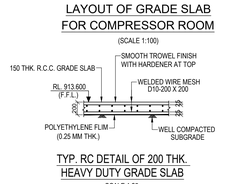
REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	30.11..2020	ISSUED FOR REVIEW	PDK	SGC	PRI	
C01	08.01..2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C01	PDK	SGC	PRI	
C02	03.03.2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C02	PDK	SGC	PRI	
C03	06.04.2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C03	AMJ	SGC	PRI	
C04	06.05.2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C04	AMJ	SGC	PRI	
C05	30.07.2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C05	SYA	SJM	AG	
C06	09.02.2022	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C06	SYA	SJM	AG	
D01	17.02.2023	AS BUILT	SYA	SJM	AG	

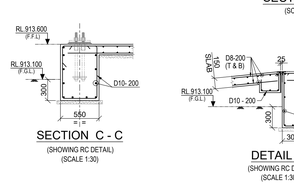




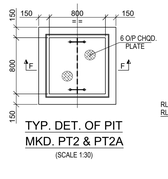
TYP. RC DETAIL OF 200 THK. HEAVY DUTY GRADE SLAB
SCALE 1:25



DETAIL - '3'
(SCALE 1:30)

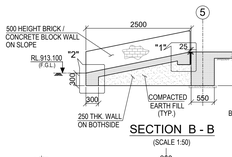


SECTION C - C
(SCALE 1:30)



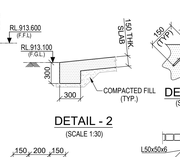
TYP. DET. OF PIT MKD. PT2 & PT2A
(SCALE 1:30)

SECTION C - C
(SHOWING RC DETAIL)
(SCALE 1:30)

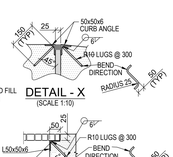


SECTION B - B
(SCALE 1:50)

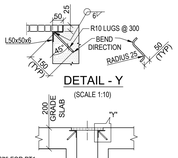
DETAIL - 1
(SCALE 1:30)



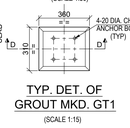
DETAIL - 2
(SCALE 1:30)



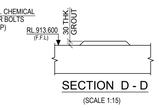
DETAIL - X
(SCALE 1:10)



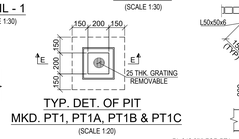
DETAIL - Y
(SCALE 1:10)



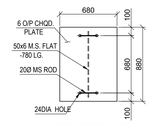
TYP. DET. OF GROUT MKD. GT1
(SCALE 1:15)



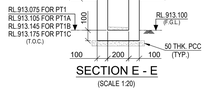
SECTION D - D
(SCALE 1:15)



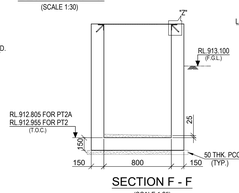
TYP. DET. OF PIT MKD. PT1, PT1A, PT1B & PT1C
(SCALE 1:20)



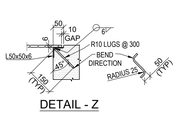
TYP. DET. OF CHQD. PLATE
(SCALE 1:25)



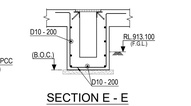
SECTION E - E
(SCALE 1:20)



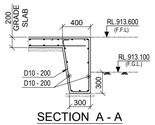
SECTION F - F
(SCALE 1:25)



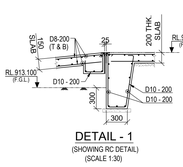
DETAIL - Z
(SCALE 1:10)



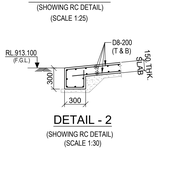
SECTION E - E
(SHOWING RC DETAIL)
(SCALE 1:25)



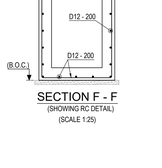
SECTION A - A
(SHOWING RC DETAIL)
(SCALE 1:30)



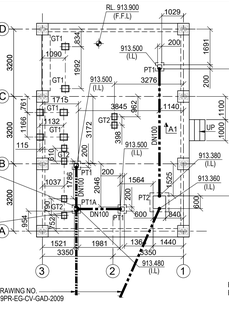
DETAIL - 1
(SHOWING RC DETAIL)
(SCALE 1:30)



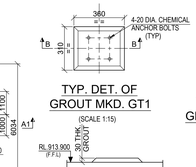
DETAIL - 2
(SHOWING RC DETAIL)
(SCALE 1:30)



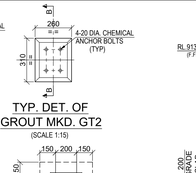
SECTION F - F
(SHOWING RC DETAIL)
(SCALE 1:25)



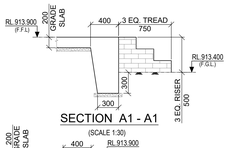
LAYOUT OF GRADE SLAB FOR PUMP SHED
(SCALE 1:100)



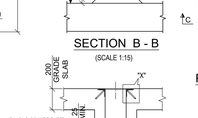
TYP. DET. OF GROUT MKD. GT1
(SCALE 1:15)



TYP. DET. OF GROUT MKD. GT2
(SCALE 1:15)



SECTION A1 - A1
(SCALE 1:30)



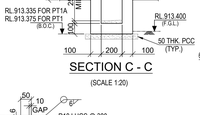
SECTION B - B
(SCALE 1:15)



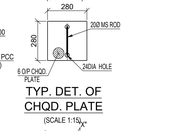
TYP. DET. OF PIT MKD. PT1 & PT1A
(SCALE 1:20)



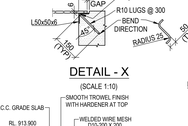
SECTION A1 - A1
(SHOWING RC DETAIL)
(SCALE 1:30)



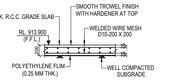
SECTION C - C
(SCALE 1:20)



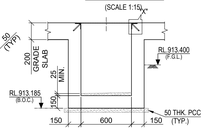
TYP. DET. OF CHQD. PLATE
(SCALE 1:10)



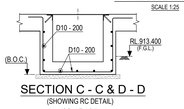
DETAIL - X
(SCALE 1:10)



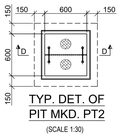
TYP. RC DETAIL OF 200 THK. HEAVY DUTY GRADE SLAB
SCALE 1:25



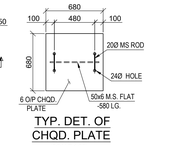
SECTION D - D
(SCALE 1:25)



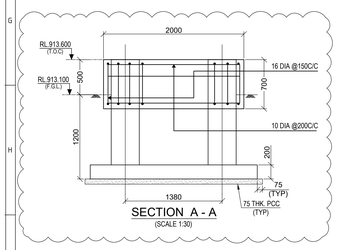
SECTION C - C & D - D
(SHOWING RC DETAIL)
(SCALE 1:25)



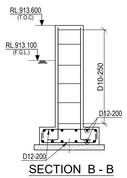
TYP. DET. OF PIT MKD. PT2
(SCALE 1:30)



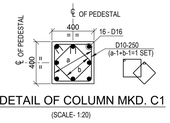
TYP. DET. OF CHQD. PLATE
(SCALE 1:25)



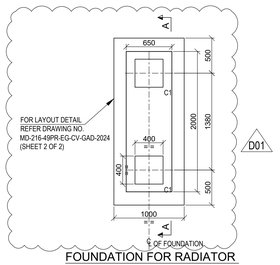
SECTION A - A
(SCALE 1:30)



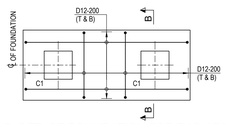
SECTION B - B
(SHOWING RC DETAIL)
(SCALE 1:30)



RC DETAIL OF COLUMN MKD. C1
(SCALE- 1:20)



FOUNDATION FOR RADIATOR PANEL MOUNTING
(6 NOS. REQUIRED)
(SCALE 1:30)



R.C. DET. OF FOUNDATION FOR RADIATOR PANEL MOUNTING
(6 NOS. REQUIRED)
(SCALE 1:30)

LEGEND :-

R.L. - REDUCED LEVEL.
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
TYP. - TYPICAL.
P.C.C - PLAIN CEMENT CONCRETE
℄ - CENTRE LINE
MKD. - MARKED
B.O.F. - BOTTOM OF FOUNDATION
THK. - THICK
FDN. - FOUNDATION
R.C. - REINFORCED CONCRETE
DIA - DIAMETER
DWG - DRAWING
C/C- CENTRE TO CENTRE
OS - OILY WATER DRAINAGE PIPE
I.L - INVERT LEVEL
CE - CONDUIT EARTHING
PS - PIPE SLEEVE
CHQD. PL - CHEQUERED PLATE.

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE 35 MPa FOR FOUNDATION & 15 MPa FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) - ASTM A615/A615M.
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 500KN/ SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE
   a) FOUNDATION = 75 mm (BOTTOM & SIDE) 50 mm. (TOP)
   b) COLUMN & PEDESTAL = 50mm
   c) BEAM = 50mm
   d) SLAB = 20mm
6. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
7. FOR ALL OTHER NOTES REFER DRAWING NO. MD-101-SMEM-EG-ST-DST-0001 TO 0008.
8. GENERAL NOTES FOR CIVIL WORKS & STANDARD REINFORCEMENT DETAIL REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001 & 1005.
9. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH REFER DRAWING NO. MD-216-49PR-EG-CV-GAD-2013 & MD-216-49PR-EG-CV-GAD-2017.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	19/02/2023	AS-BUILT & REVISED AS MKD. 001	SYA	SJM	AVG	AG	
		C06	09/02/2022	ISSUED FOR CONSTRUCTION & REVISED AS MKD. 002	SYA	SJM	AVG	AG	
		C05	30/07/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. 002	SYA	SJM	AVG	AG	
		C04	06/05/2021	ISSUED FOR CONSTRUCTION 002	AMJ	SGC	SHD	PRI	
		C03	06/04/2021	ISSUED FOR CONSTRUCTION 002	AMJ	SGC	SHD	PRI	
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS	C02	03/03/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. 002	PDK	SGC	SHD	PRI	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL - DESIGN CRITERIA	C01	08/01/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. 002	PDK	SGC	SHD	PRI	
MD-216-49PR-EG-ME-LAD-V553-1002	LAYOUT OF POWER GENERATION	A01	30/11/2020	ISSUED FOR REVIEW	AMJ/PDK	SGC	SHD	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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MA'ADEN معادن
Saudi Arabian Mining Company شركة المعادن العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER: 019045_008_C_G_24298_006 — REVISION: 7
OUTOTEC DRAWING NUMBER: 900707__GAU01__WG01__MTL01__45092 — DOCUMENT ID: OU100690015 — REVISION: 7

DRAWN	AMJ	10/11/2020
CHECKED	SGC	20/11/2020
ENGINEER	SHD	08/11/2020
APPROVED	PRI	30/11/2020
EMPLOYER		
SCALE	1:200,100,20	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
POWER PLANT MISCELLANEOUS STRUCTURE
GA AND DETAIL (SHEET 1 OF 2)

DRAWING NUMBER: MD-216-49PR-EG-CV-GAD-2024 — REV: D01

DRAWING NUMBER: MD-216-49PR-EG-CV-GAD-2024 | REV: D01

COMPACT WORKSHOP & WAREHOUSE

DAY TANK AREA

FUTURE RADIATOR EXTENSION

FUTURE ENGINE HALL EXTENSION

ENGINE HALL

ELECTRICAL SUBSTATION

RADIATOR SUPPORTING STRUCTURE

RADIATOR PANEL MOUNTING FOUNDATION

BOOSTER ROOM-1

BOOSTER ROOM-2

E=263907.032
N=2506023.652

E=263910.954
N=2506024.924

E=263961.291
N=2506057.241

E=263964.789
N=2506053.531

E=263968.849
N=2506046.882

E=263974.128
N=2506049.002

E=263969.506
N=2505993.918

E=264005.033
N=2506068.121

E=264000.570
N=2506041.129

E=264004.333
N=2506036.994

E=264019.080
N=2506021.713

ALCO COLLECTING PIT
BOTTOM OF PIT (910.300)

ALCO COLLECTING PIT
BOTTOM OF PIT (910.920)

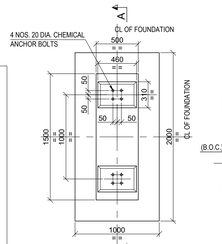
EXISTING OIL WATER COLLECTING SUMP TO BE REMOVE FRDM (DRG. NO. MD-216-49PR-EG-CV-GAD-1596)

DUCT SUPPORT
REFER DRAWING NO. MD-216-49PR-EG-CV-GAD-1602)

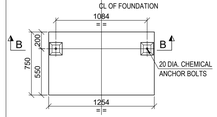
DN150 HDPE
FROM COMPRESSOR ROOM PIT
MD-216-49PQ-EG-CV-GAD-2024
(SHEET 1 OF 2)

49PR/10 49PR/9 49PR/8 49PR/7 49PR/6 49PR/5 49PR/4 49PR/3 49PR/2 49PR/1
49PR/A 49PR/B 49PR/C D01

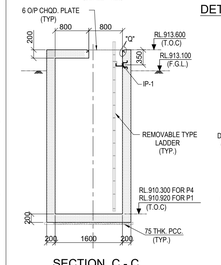
LAYOUT PLAN
(SCALE 1:150)



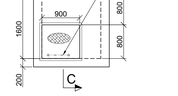
DETAIL OF FOUNDATION MKD. SF1
(FOUNDATION MKD. SF2 SIMILAR)
(ORIENTATION AS PER LAYOUT PLAN)
(SCALE - 1:25)



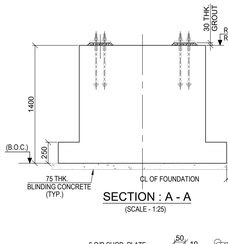
DETAIL OF PEDESTAL MKD. P1
(PEDESTAL MKD. P2 SIMILAR)
(SCALE - 1:25)



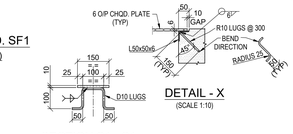
SECTION C - C
(SCALE 1:50)



TYP. DETAIL OF PIT MKD. PT1 & PT4
(ORIENTATION AS PER LAYOUT PLAN)
(SCALE 1:50)

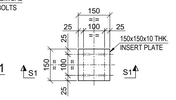


SECTION : A - A
(SCALE - 1:25)

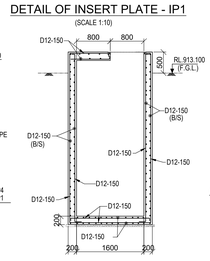


DETAIL - X
(SCALE 1:10)

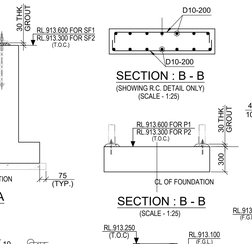
SECTION S1 - S1
(SCALE 1:10)



DETAIL OF INSERT PLATE - IP1
(SCALE 1:10)

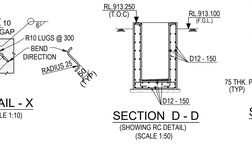


SECTION C - C
(SHOWING RC DETAIL)
(SCALE 1:50)

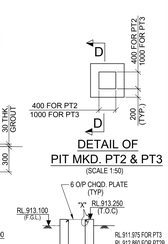


SECTION : B - B
(SHOWING R.C. DETAIL ONLY)
(SCALE - 1:25)

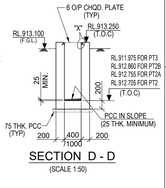
SECTION : B - B
(SCALE - 1:25)



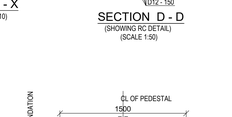
SECTION D - D
(SHOWING RC DETAIL)
(SCALE 1:50)



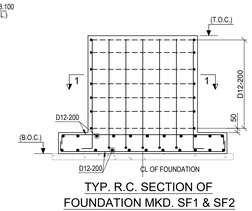
DETAIL OF PIT MKD. PT2 & PT3
(SCALE 1:50)



SECTION D - D
(SCALE 1:50)



SECTION : 1 - 1
(SCALE - 1:25)



TYP. R.C. SECTION OF FOUNDATION MKD. SF1 & SF2
(SCALE - 1:25)

LEGEND :-

R.L - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
F.G.L. - FINISHED GROUND LEVEL
TYP. - TYPICAL
P.C.C - PLAIN CEMENT CONCRETE
℄ - CENTRE LINE
MKD. - MARKED
B.O.F. - BOTTOM OF FOUNDATION
THK. - THICK
FDN. - FOUNDATION
R.C. - REINFORCED CONCRETE
DIA - DIAMETER
DWG - DRAWING
C/C- CENTRE TO CENTRE
OS. - OILY WATER DRAINAGE PIPE
I.L. - INVERT LEVEL
CE - CONDUIT EARTHING
PS - PIPE SLEEVE

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE 35 MPa FOR FOUNDATION & 15 MPa FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) - ASTM A615/A615M.
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 500KN/ SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE
   a) FOUNDATION = 75 mm (BOTTOM & SIDE) 50 mm. (TOP)
   b) COLUMN & PEDESTAL = 50mm
   c) BEAM = 50mm
   d) SLAB = 20mm
6. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
7. FOR ALL OTHER NOTES REFER DRAWING NO. MD-101-SMEM-EG-ST-DST-0001 TO 0008.
8. GENERAL NOTES FOR CIVIL WORKS & STANDARD REINFORCEMENT DETAIL REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001 & 1005.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	17/02/2023	AS-BUILT & REVISED AS MKD. (01)	SYA	SJM	AVG	AG	
		C06	09/02/2022	ISSUED FOR CONSTRUCTION & REVISED AS MKD.	SYA	SJM	AVG	AG	
		C05	30/07/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. (02)	SYA	SJM	AVG	AG	
		C04	06/05/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. (03)	AMJ	SGC	SHD	PRI	
		C03	06/04/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. (04)	AMJ	SGC	SHD	PRI	
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	C02	03/03/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. (05)	PDK	SGC	SHD	PRI	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL - DESIGN CRITERIA	C01	08/01/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. (06)	PDK	SGC	SHD	PRI	
MD-216-49PR-EG-ME-LAD-Y553-1002	LAYOUT OF POWER GENERATION	A01	30/11/2020	ISSUED FOR REVIEW	AMJ/PDK	SGC	SHD	PRI	

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REVISION	7
OUTOTEC DRAWING NUMBER	900707__GAU01__WG01__MTL01__45092
DOCUMENT ID	OU100690015
REVISION	7
DRAWN	AMJ 10/11/2020
CHECKED	SGC 20/11/2020
ENGINEER	SHD 08/11/2020
APPROVED	PRI 30/11/2020
EMPLOYER	
SCALE	1:200,100,20
	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
POWER PLANT MISCELLANEOUS STRUCTURE
GA AND DETAIL (SHEET 2 OF 2)

DRAWING NUMBER: MD - 216 - 49PR - EG - CV - GAD - 2024 | REV: D01